

Haotian Zeng

haotianswebsite.com

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Go, Python, Java, Bash, HTML/CSS/JavaScript, Ruby, C, C++, Groovy
PostgreSQL, gRPC, Terraform, Prometheus, AWS, Azure, Docker, Jenkins

EXPERIENCE

Hashicorp – Backend Engineer (Boundary)

Dec 2021 – June 2023

- Designed and implemented Metrics Package, providing Cloud Team insight into thousands of live clusters, facilitating real-time diagnosis and helping ensure 99.9% uptime for customer clusters, enabling the company to uphold its SLA commitments and enhance revenue streams.
- Designed and implemented Vault SSH Certificate Injection feature, providing a more convenient workflow for authenticating session credentials, streamlining user interaction by 20%.
- Refactored PostgreSQL database, programmatically changing thousands of table and constraint names to follow established convention, improving scalability and consistency of codebase.
- Designed and implemented Controller Metadata functionality, delivering license metadata to a Front-End API endpoint, providing essential data to determine versioning of Enterprise product.
- Captained releases by coordinating release candidate testing, conducted local testing for new features, and wrote testcases in the codebase, ensuring the quality of software at all stages.

Lawrence Berkeley National Laboratory – Research Assistant

June 2021 – Nov 2021

- Designed intuitive wrapper using MVC principles to seamlessly launch geochemistry simulation software with complex input parameters and receive results as plottable data.
- Employed sensitivity analysis and machine learning optimization algorithms on diverse datasets, enabling my lab group to publish a peer-reviewed manuscript in a scientific journal.

Wind River Systems – Software Engineering Intern

June 2019 – Dec 2019

- Designed and implemented CI tool to specify complex testcases and parameters, providing an important CI tool for engineers, increasing testing pipeline speed by up to 6 hours per run.
- Implemented query tool for Sales team, revitalizing access to an underutilized database and enabling sales opportunities.

PROJECTS

HaotiansWebsite

Sept 2024 - current

- Designed and developed personal website using HTML/CSS/JS to showcase professional portfolio.
- Implemented principles such as responsive design, image preloading, url obfuscation, and “captcha” verification, learning good design principles and acquiring new skills as needed.

SSH Certificate Injection (Hashicorp)

Dec 2022 – Jan 2023

- Created first major Boundary Enterprise feature enabling direct injection of new SSH keys into Vault.
- Restructured and improved user flow, providing a 20% quicker and more secure experience when authorizing a Boundary session connection.
- Collaborated in small team to split complex feature into manageable parts, resulting in the successful launch of a major Boundary enterprise feature within the financial quarter.

Boundary Metrics (Hashicorp)

Feb 2022 – March 2022

- Wrote server and client-side metrics to capture latency, request size, connection count, and connection status for gRPC, HTTP, and WebSocket protocol data, to a specified endpoint.
- Led the initial design document writing process, consulted with senior engineers, read and created documentation to swiftly gain domain expertise and advance the development process.
- Refactored package and added tests, documentation, and new metrics for additional connection types, taking on responsibility beyond the initial ask, as requirements evolved.

Monte Carlo Uncertainty Quantifier (LBNL)

June 2021 – Nov 2021

- Analyzed the effect of different soil geochemical compositions on uranium seepage in groundwater by programmatically triggering multiple instances of PHREEQC simulation software and processing results with the Pandas and Pyplot Python libraries.

Dragonfly Pipeline (Wind River)

June 2019 – Dec 2019

- Integrated Jenkins, Python, Bash, and Groovy scripts to develop a robust CI tool, streamlining the testing workflow for core product development, while delivering run logs, errors, and notifications.
- Implemented cron jobs to update dynamic selection menu with valid branching testcases and options.

EDUCATION

University of California, Berkeley – Computer Science, Nuclear Engineering

Relevant Courses: Data Structures, Computer Architecture, Probability, Computer Security, Operating Systems & Systems Programming, Artificial Intelligence, Linear Algebra, Multivariable Calculus